

Practical Regularized Quasi-Newton Methods with Inexact Function Values

Hiroki Hamaguchi  · Naoki Marumo  ·
Akiko Takeda 

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Abstract Many practical optimization problems involve objective function values that are corrupted by unavoidable numerical errors. In smooth nonconvex optimization, quasi-Newton methods combined with line search are widely used due to their efficiency and scalability. These methods implicitly assume accurate function evaluations and thus may fail to converge in noisy settings. Developing fast and stable quasi-Newton methods for such scenarios is therefore crucial.

To address this issue, we propose a noise-tolerant regularized quasi-Newton method equipped with a relaxed Armijo-type line search, designed to remain stable under inaccurate function evaluations. By combining a regularization parameter update rule inspired by Objective-Function-Free Optimization and the AdaGrad-Norm method, **we prove an order-optimal $\mathcal{O}(1/\varepsilon^2)$ complexity bound under errors in function values.**

We performed extensive experiments on the CUTEst benchmark collection with artificially noisy objective function evaluations, as well as with low-precision floating-point arithmetic (64-, 32-, and 16-bit). The results demonstrate **stable performance under reduced precision and function-value noise** while retaining practical convergence speed and computational cost.

Keywords Regularized Quasi-Newton Method · Numerical Stability · Numerical Error · AdaGrad-Norm · Objective-Function-Free Optimization

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H. Hamaguchi

Graduate School of Information Science and Technology, The University of Tokyo, Tokyo, Japan
E-mail: hirokihamaguchi0331@gmail.com

N. Marumo

Graduate School of Information Science and Technology, The University of Tokyo, Tokyo, Japan

A. Takeda

Graduate School of Information Science and Technology, The University of Tokyo, Tokyo, Japan
Center for Advanced Intelligence Project, RIKEN, Tokyo, Japan

1 Introduction

We consider the unconstrained optimization problem

$$\underset{x \in \mathbb{R}^n}{\text{minimize}} \quad f(x), \quad (1)$$

where $f: \mathbb{R}^n \rightarrow \mathbb{R}$ is a continuously differentiable nonconvex function, and $x \in \mathbb{R}^n$ is the optimization variable. We focus on settings where objective function evaluations are contaminated by bounded and non-diminishing numerical noise [38]. Specifically, only inexact function values $\bar{f}(x)$ are available for optimization. Such inaccuracies are unavoidable in many realistic scenarios, including finite-precision arithmetic, simulation-based evaluations, and stochastic approximations [4, 29, 37]. When function values are contaminated by numerical noise, standard optimization algorithms may lose their theoretical guarantees, behave erratically, or terminate prematurely with unreliable solutions.

In the absence of noise and with accurate function evaluations, quasi-Newton methods, in particular the Broyden–Fletcher–Goldfarb–Shanno (BFGS) method and its Limited-memory variant (L-BFGS), are among the most efficient and widely used algorithms for solving problem (1) [25, 31]. Their success is largely attributed to fast local convergence and scalability. Standard implementations [32, 41] rely on line search procedures that enforce the Wolfe conditions to guarantee sufficient descent and the positive definiteness of the Hessian approximation.

In the presence of errors in function values, the sufficient-decrease test used in these line searches may become unreliable. Differences in function values can be dominated by numerical errors, leading to unstable step sizes, ill-conditioned Hessian approximations, or abnormal termination [33, 37]. From a computational perspective, improving the stability of quasi-Newton methods in noisy environments is therefore crucial. In particular, it is desirable to design algorithms that (i) retain computational efficiency when function evaluations are accurate, (ii) remain stable when function values are unreliable, and (iii) guarantee convergence to first-order stationary points.

In this work, we propose a noise-tolerant regularized quasi-Newton method that addresses these challenges. Our approach combines an Armijo line search, quadratic regularization [22, 40], and an Objective-Function-Free Optimization (OFFO) inspired strategy [17, 19]. When a sufficient decrease in function value is observed, the method behaves similarly to standard quasi-Newton methods with line search, preserving its practical efficiency. When numerical noise dominates, we introduce regularization on the approximate Hessian to ensure numerical stability, which yields convergence properties similar to those of OFFO methods, particularly AdaGrad-Norm [43]. The resulting algorithm is particularly well-suited for low-precision or noisy environments. Extensive numerical experiments on the CUTEst benchmark collection, across various floating-point precisions and artificial noise settings, show stable performance under function-value noise and reduced precision.

The key methodological advance is to switch between unregularized quasi-Newton steps and regularized steps using an OFFO-based cumulative-gradient scale. Using this scale at every iteration avoids reliance on measured function decreases when controlling regularization but can be unnecessarily conservative when function values are reliable. Conversely, merely adding a nonvanishing error allowance to a line

Method	Uses f	Noisy f	Noisy ∇f	Nonconvex	$\mathcal{O}(\varepsilon^{-2})$ guarantee
L-BFGS [25, 32, 41]	✓	–	–	✓	–
Regularized Newton method [39]	✓	–	–	✓	✓
Regularized L-BFGS method [22]	✓	–	–	✓	–
Noise-tolerant quasi-Newton method [37, 44]	✓	✓	✓	–	–
OFFO method [17]	–	✓	–	✓	✓
Proposed method Algorithm 1	✓	✓	–*	✓	✓

Table 1: Theoretical comparison of the proposed method and related Newton-type, quasi-Newton, and OFFO methods. A checkmark (✓) denotes theoretical coverage and “–” its absence. The “Noisy f ,” “Noisy ∇f ,” and “Nonconvex” columns indicate whether the corresponding convergence theory allows function-value errors, allows gradient errors, and covers nonconvex objectives, respectively. The asterisk indicates that gradient noise is considered in numerical stress tests, but its analysis is outside the scope of this paper.

search does not guarantee convergence to arbitrarily small first-order tolerances because its accumulated contribution need not be bounded. Our test permits an unregularized step only when error-adjusted function values certify progress and otherwise activates the OFFO-based scale. The test bounds the accumulated allowances over unregularized iterations, while the cumulative-gradient scale controls the remaining iterations without requiring measured function decreases. This combination retains unregularized quasi-Newton steps and establishes an $\mathcal{O}(1/\varepsilon^2)$ worst-case iteration complexity under persistent function-value errors, matching the optimal worst-case order for smooth nonconvex problems with Lipschitz-continuous gradients [5].

1.1 Comparison With Related Work

We next compare our method with related quasi-Newton and OFFO methods, whose theoretical guarantees are summarized in Table 1.

The closest methods are the regularized L-BFGS method [22] and Noise Tolerant Quasi-Newton (NTQN) [37]. The regularized L-BFGS method assumes reliable function values and establishes global convergence to first-order stationarity [22, Theorem 2]. Shi et al. [37] develop NTQN for bounded errors in both function and gradient evaluations, using a noise-tolerant line search and a lengthening procedure to stabilize the quasi-Newton updates. Their analysis allows bounded gradient errors and establishes convergence to a noise-dependent neighborhood for strongly convex objectives [37, Theorem 3.5]. Our relaxed Armijo test is related to their noise-tolerant test [37, Section 4], whereas our switch controls the accumulated error allowances

and yields an $\mathcal{O}(1/\varepsilon^2)$ bound for arbitrarily small first-order tolerances on smooth nonconvex objectives with accurate gradients.

Regularized Newton and OFFO methods provide further theoretical context [17, 19, 39]. Regularized Newton methods attain an $\mathcal{O}(1/\varepsilon^2)$ iteration bound with reliable function values [39, Theorem 4.1], while OFFO methods avoid function values and attain the same order for smooth nonconvex objectives [17, 19]. Our method instead uses inexact function values to permit unregularized quasi-Newton steps when progress is certified and retains this worst-case order under persistent function-value errors.

1.2 Organization of the Paper

The remainder of this paper is organized as follows. In section 2, we review relevant background and introduce the problem setting. In section 3, we present the proposed algorithm. In section 4, we establish a global convergence property of the proposed algorithm. In section 5, we discuss practical choices of algorithmic variables and parameters. In section 6, we report numerical results on the CUTEst benchmark collection. Finally, in section 7, we conclude the paper with a summary and future research directions.

Our implementation is publicly available on GitHub¹.

2 Preliminaries and Problem Setting

We begin by reviewing relevant background and formalizing the problem setting.

2.1 Regularized Quasi-Newton Method

A regularized quasi-Newton method is a variant of quasi-Newton methods that does not rely heavily on line-search techniques. This property is important for optimization in noisy evaluation environments. There has been renewed interest in regularized Newton-type methods, including cubic-regularization schemes [30, 34, 42], and we focus on quadratic regularization [9, 22, 24, 36, 39, 40, 46] for simplicity of analysis and implementation.

For $g_k := \nabla f(x_k)$ and an approximate Hessian $B_k \in \mathbb{R}^{n \times n}$, regularized quasi-Newton methods compute the step as

$$x_{k+1} = x_k + d_k, \quad d_k = -(B_k + \mu_k I)^{-1} g_k,$$

with a regularization parameter $\mu_k \geq 0$. Combining regularization with line search is also possible and has been explored [39]. Several strategies for choosing μ_k in nonconvex optimization have been proposed [36, 39, 40], and one of them is given by

$$\mu_k = c_1 \max(0, -\lambda_{\min}(B_k)) + c_2 \|\nabla f(x_k)\|^\delta,$$

¹ <https://github.com/HirokiHamaguchi/qnlab>

where $c_1 > 1$, $c_2 > 0$, $\delta \geq 0$, and $\lambda_{\min}(B_k)$ denotes the smallest eigenvalue of B_k . This choice guarantees that $B_k + \mu_k I$ is positive definite, ensuring that d_k is a descent direction. We adapt such regularization strategies to noisy evaluation environments.

2.2 Objective-Function-Free Optimization

Objective-Function-Free Optimization (OFFO) refers to a class of methods that avoid explicit evaluations of the objective function and rely solely on gradient information [17, 19]. Such methods are particularly attractive when function values are unreliable due to noise. Similar adaptive trust-region approaches are also discussed [16].

Prominent examples of OFFO include adaptive gradient methods such as AdaGrad [11] and AdaGrad-Norm [43]. Their update rules are

$$\text{(AdaGrad)} \quad x_{k+1}^{(i)} = x_k^{(i)} - \frac{1}{\theta \sqrt{\varsigma + \sum_{j=0}^k (g_j^{(i)})^2}} g_k^{(i)}, \quad (i = 1, 2, \dots, n)$$

$$\text{(AdaGrad-Norm)} \quad x_{k+1} = x_k - \frac{1}{\theta \sqrt{\varsigma + \sum_{j=0}^k \|g_j\|^2}} g_k$$

respectively, where $\varsigma > 0$ and $\theta > 0$ are algorithmic parameters, and $x^{(i)}$ denotes the i -th component of a vector x .

2.3 Problem Setting

We consider optimization problems in which exact objective function values are unavailable, following the recent literature on error-aware optimization [1, 6, 7, 18, 35, 38]. When both objective values and gradients are heavily corrupted by noise, reliable optimization cannot be guaranteed. Thus, meaningful optimization requires that at least one source of information be sufficiently accurate. This work focuses on problems arising from finite-precision computations or inexact evaluations, where objective function evaluations are subject to non-negligible errors while gradient information can be computed with relatively high accuracy. Settings dominated by gradient errors, such as stochastic approximation and inexact-gradient methods [2, 37, 44], are beyond the scope of this work.

In the following, we formalize the assumptions on function value and gradient evaluations. We assume that only an error-contaminated function value $\bar{f}(x)$ is accessible, satisfying the hybrid absolute-relative error model [20, 21]:

$$|f(x) - \bar{f}(x)| \leq \epsilon_f \max(1, |f(x)|), \quad (2)$$

where $0 \leq \epsilon_f < 1$ is an error rate. The error is predominantly relative when $|f(x)|$ is large and effectively absolute when $|f(x)|$ is small, capturing the typical behavior of rounding errors in IEEE 754 floating-point arithmetic [20]. The value of ϵ_f is a supplied safety margin for errors in the computed function values, with its practical specification and sensitivity to misspecification discussed in section 6.

We next specify the assumption on gradient evaluations. Let $g(x)$ be the gradient returned by the available gradient oracle at x . In practice, termination is typically based on a gradient tolerance ϵ_{gtol} , and $\|g(x)\| \leq \epsilon_{\text{gtol}}$ is used as a stopping criterion. The convergence analysis adopts the following exact-gradient assumption.

Assumption 1 (Exact gradient) *The algorithm has access to a gradient oracle g such that, for every evaluated point x ,*

$$g(x) = \nabla f(x). \quad (3)$$

The experiments using only function-value noise satisfy Assumption 1. Experiments with joint noise or reduced precision complement the theory by evaluating empirical performance beyond the exact-gradient setting.

3 Proposed Algorithm

We propose a new algorithm for the problem setting stated in section 2.3. We draw inspiration from the OFFO framework to design robust regularization and scaling strategies. Unlike purely OFFO methods, our algorithm exploits function values when they are numerically reliable and switches to gradient-based control otherwise. The test in line 3 permits an unregularized quasi-Newton step when the error-adjusted function values certify progress and otherwise invokes the OFFO-based scale. Function values are used in the line search in both modes, whereas the OFFO-based regularization scale depends only on gradient information. The pseudocode of our algorithm is given in Algorithm 1. The minimum of the empty set is defined as $+\infty$ in line 3. Algorithm 2 is an auxiliary line search algorithm called in Algorithm 1. In the following, we explain the three main components of Algorithm 1.

Algorithm 1: Noise Tolerant Regularized Quasi-Newton Method

Input: $x_0, 0 < k_{\max}, 0 \leq \epsilon_f < 1, 0 < \theta_{\min} \leq \theta_{\max}$ and $0 < \varsigma < 1$

- 1 $k \leftarrow 0, g_0 \leftarrow g(x_0), K^0 \leftarrow \emptyset, K^+ \leftarrow \emptyset$
- 2 **for** $k = 0, 1, 2, \dots, k_{\max} - 1$ **do**
- 3 **if** $\min_{j \in K^0} (\bar{f}(x_j) - \Delta_j) \geq \bar{f}(x_k)$ **then**
- 4 $K^0 \leftarrow K^0 \cup \{k\}$
- 5 $\mu_k \leftarrow 0$
- 6 **else**
- 7 $K^+ \leftarrow K^+ \cup \{k\}$
- 8 $\mu_k \leftarrow \theta_k \sqrt{\varsigma + \sum_{j \in K^+} \|g_j\|^2}$ with $\theta_k \in [\theta_{\min}, \theta_{\max}]$ (see section 5.3).
- 9 $d_k \leftarrow -(B_k + \mu_k I)^{-1} g_k$ with B_k (see section 5.1).
- 10 Find α_k and Δ_k satisfying eq. (4) by Algorithm 2.
- 11 $x_{k+1} \leftarrow x_k + \alpha_k d_k, g_{k+1} \leftarrow g(x_{k+1})$
- 12 **end**

Algorithm 2: Backtracking Line Search with Relaxed Armijo Condition

Input: $x_k \in \mathbb{R}^n$, $d_k \in \mathbb{R}^n$, $g_k \in \mathbb{R}^n$, $0 < c < 1$ and $0 < \beta_{\min} < \beta_{\max} < 1$

- 1 $\alpha_k \leftarrow 1$
- 2 $\Delta_k \leftarrow \frac{2\varepsilon_f}{1-\varepsilon_f} \max(1, \bar{f}(x_k), -\bar{f}(x_k + \alpha_k d_k))$
- 3 **while** $\bar{f}(x_k) + c\alpha_k g_k^\top d_k + \Delta_k < \bar{f}(x_k + \alpha_k d_k)$ **do**
- 4 $\alpha_k \leftarrow \beta \alpha_k$ with $\beta \in [\beta_{\min}, \beta_{\max}]$ (see section 5.2).
- 5 $\Delta_k \leftarrow \frac{2\varepsilon_f}{1-\varepsilon_f} \max(1, \bar{f}(x_k), -\bar{f}(x_k + \alpha_k d_k))$
- 6 **end**
- 7 **return** α_k, Δ_k

Computation of the Regularized Quasi-Newton Direction (line 9): The first component is the computation of the regularized quasi-Newton direction d_k stated in section 2.1. Given $g_k = g(x_k) = \nabla f(x_k)$, an approximate Hessian B_k , and a regularization parameter μ_k , we compute the search direction $d_k = -(B_k + \mu_k I)^{-1} g_k$. A practical choice of B_k is discussed in section 5.1.

Relaxed Armijo Condition with an Error-Absorbing Term (line 10): The second component is the computation of the step size α_k using a relaxed Armijo condition, which is defined as follows:

$$\begin{cases} \bar{f}(x_k) + c\alpha_k g_k^\top d_k + \Delta_k \geq \bar{f}(x_k + \alpha_k d_k), \\ \Delta_k = \frac{2\varepsilon_f}{1-\varepsilon_f} \max(1, \bar{f}(x_k), -\bar{f}(x_k + \alpha_k d_k)), \end{cases} \quad (4)$$

where Δ_k is the error-absorbing term that is recomputed for each α_k . As shown in the proof of Lemma 1, Δ_k bounds the absolute discrepancy between the true and observed function decreases whenever the true function value decreases. This relaxation guarantees the existence of $\alpha_k \in (0, 1]$ even in the presence of errors in $\bar{f}$. When d_k is a descent direction ($g_k^\top d_k < 0$), eq. (4) ensures that a temporary increase of $\bar{f}$ is at most Δ_k . Since Δ_k depends on $-\bar{f}(x_k + \alpha_k d_k)$ but not on $\bar{f}(x_k + \alpha_k d_k)$, Δ_k does not increase when $\bar{f}(x_k + \alpha_k d_k)$ becomes large. This prevents unbounded growth of Δ_k .

Adaptive Update of the Regularization Parameter (lines 3–8): The third component is the adaptive update of the regularization parameter μ_k . A larger μ_k yields a more conservative step, while a smaller μ_k allows a more aggressive quasi-Newton step, especially when $\mu_k = 0$. In practical settings, $\mu_k = 0$ is preferred, since this allows us to fully exploit the quasi-Newton approximation and typically leads to practically faster convergence. Let us partition the iteration indices $K := \{0, 1, 2, \dots, k_{\max} - 1\}$ into two sets:

$$K^0 := \{k \in K \mid \mu_k = 0\}, \quad K^+ := \{k \in K \mid \mu_k > 0\}. \quad (5)$$

We set $\mu_k = 0$ if and only if the following condition is satisfied:

$$\min_{j \in K^0, j < k} (\bar{f}(x_j) - \Delta_j) \geq \bar{f}(x_k). \quad (6)$$

In Algorithm 1, K^0 is updated incrementally and therefore contains only indices $j < k$ when the test in line 3 is evaluated, making the notation in that line consistent with eq. (6). Note that $k = 0$ satisfies eq. (6) since K^0 is empty. This condition restricts $\mu_k = 0$ to iterations for which a sufficient decrease in $\bar{f}$ has been observed. Thus, although $\bar{f}$ can temporarily increase, every iteration in K^0 must improve the previous error-adjusted reference value. Otherwise, to ensure convergence to first-order stationary points, we apply an OFFO-based update

$$\mu_k = \theta_k \sqrt{\varsigma + \sum_{j \in K^+, j \leq k} \|g_j\|^2}, \quad (7)$$

as described in section 2.2. Since OFFO schemes do not rely on function values, this update is robust to numerical errors in $\bar{f}$.

The rationale for why a purely regularized quasi-Newton method is inadequate, thus necessitating the additional OFFO-inspired fallback mechanism, is that noisy function-value comparisons cannot reliably determine when regularization is needed. The underlying issue is the standard globalization problem that a full Newton-type step based on an inaccurate Hessian approximation can be too long and is normally safeguarded by a line search [31, Section 3.1]. Let $f(x) = \bar{f}(x) = x^2/2$, $B_k = 1/2$, and $x_0 = a > 0$. The unregularized full step maps a to $-a$, so repeated full steps cycle between these two points without reducing the function value or the nonzero gradient norm. An ordinary Armijo line search with accurate function values rejects this step because it produces no decrease. With a positive supplied error allowance, however, the relaxed Armijo condition accepts the step for sufficiently small $a > 0$ because Δ_k masks the zero decrease. At the following iteration, the switch in eq. (6) detects that no error-adjusted progress has been certified and activates the OFFO-inspired scale before a second unregularized step is taken. Thus, the fallback provides function-value-independent regularization precisely when the relaxed line search cannot distinguish progress from evaluation error.

4 Theoretical Analysis

In this section, we establish the global convergence properties of Algorithm 1 under the problem setting in section 2.3, namely, eqs. (2) and (3).

4.1 Assumptions

In addition to the exact-gradient assumption stated in Assumption 1, we make the following three structural assumptions. The first assumption ensures that the objective function is bounded below.

Assumption 2 *The function f is bounded below, meaning that for all $x \in \mathbb{R}^n$,*

$$f(x) \geq f^* := \inf_{x \in \mathbb{R}^n} f(x) > -\infty.$$

By the triangle inequality and eq. (2), we have $\bar{f}(x) \geq f(x) - \varepsilon_f \max(1, |f(x)|)$. Since the function $g(t) = t - \varepsilon_f \max(1, |t|)$ is non-decreasing for $0 \leq \varepsilon_f < 1$, it follows that $\inf_x g(f(x)) = g(f^*)$. Consequently, we can also lower bound $\bar{f}$ as

$$\bar{f}(x) \geq \bar{f}^* := \inf_{x \in \mathbb{R}^n} (f(x) - \varepsilon_f \max(1, |f(x)|)) = f^* - \varepsilon_f \max(1, |f^*|) > -\infty. \quad (8)$$

The second assumption ensures smoothness of the objective function.

Assumption 3 *The function $f: \mathbb{R}^n \rightarrow \mathbb{R}$ is L -smooth, meaning that there exists a constant $L > 0$ such that for all $x, y \in \mathbb{R}^n$,*

$$f(y) \leq f(x) + \nabla f(x)^\top (y - x) + \frac{L}{2} \|y - x\|^2.$$

The third assumption requires the matrices $\{B_k\}_{k=0}^{k_{\max}-1}$ to be uniformly positive definite and bounded.

Assumption 4 *There exist constants $0 < m \leq M$ such that all B_k satisfy*

$$mI \preceq B_k \preceq MI.$$

As discussed in section 5, the shifted raw L-BFGS construction with selected curvature pairs guarantees Assumption 4. By Assumption 4, we can derive useful bounds involving g_k and d_k . Using $d_k = -(B_k + \mu_k I)^{-1} g_k$ (line 9) and Assumption 4 asserting $(m + \mu_k)I \preceq B_k + \mu_k I \preceq (M + \mu_k)I$, we obtain the following bounds:

$$-g_k^\top d_k = d_k^\top (B_k + \mu_k I) d_k \geq (m + \mu_k) \|d_k\|^2 \geq m \|d_k\|^2, \quad (9)$$

$$-g_k^\top d_k = g_k^\top (B_k + \mu_k I)^{-1} g_k \geq \frac{1}{M + \mu_k} \|g_k\|^2, \quad (10)$$

$$\|d_k\|^2 = \|(B_k + \mu_k I)^{-1} g_k\|^2 \leq \frac{1}{(m + \mu_k)^2} \|g_k\|^2. \quad (11)$$

4.2 Error Bound on Function Evaluations

We first establish useful properties of the error-absorbing term Δ_k defined in eq. (4). In the following, we define the actual and observed function decreases as

$$\delta_k := f(x_k) - f(x_{k+1}), \quad (12)$$

$$\bar{\delta}_k := \bar{f}(x_k) - \bar{f}(x_{k+1}). \quad (13)$$

The first property bounds the difference between the actual decrease δ_k and the observed decrease $\bar{\delta}_k$ when the true function value decreases.

Lemma 1 *If x_k and x_{k+1} satisfy $f(x_{k+1}) \leq f(x_k)$, it holds that*

$$\delta_k \leq \bar{\delta}_k + \Delta_k.$$

Proof We start from general observations. We first prove that, for all $x \in \mathbb{R}^n$,

$$\begin{cases} \max(1, +f(x)) \leq \frac{1}{1-\varepsilon_f} \max(1, +\bar{f}(x)), \\ \max(1, -f(x)) \leq \frac{1}{1-\varepsilon_f} \max(1, -\bar{f}(x)). \end{cases} \quad (14)$$

We prove the inequality with the plus sign. If $+f(x) \leq 1$, then $\max(1, +f(x)) = 1 \leq \frac{1}{1-\varepsilon_f} \leq \frac{1}{1-\varepsilon_f} \max(1, +\bar{f}(x))$. If $+f(x) > 1$, eq. (2) implies

$$|f(x) - \bar{f}(x)| \leq \varepsilon_f \max(1, |f(x)|) = +\varepsilon_f f(x) \quad (15)$$

and thus $f(x) - \bar{f}(x) \leq +\varepsilon_f f(x)$. Hence,

$$\max(1, +f(x)) = +f(x) \leq +\frac{1}{1-\varepsilon_f} \bar{f}(x) \leq \frac{1}{1-\varepsilon_f} \max(1, +\bar{f}(x)).$$

The case with the negative sign follows analogously, using eq. (2) to obtain $-f(x) + \bar{f}(x) \leq -\varepsilon_f f(x)$ when $f(x) < -1$. This completes the proof of eq. (14).

We also have the following general bound for all $k \in K$:

$$\begin{aligned} |\delta_k - \bar{\delta}_k| &= |(f(x_k) - \bar{f}(x_k)) - (f(x_{k+1}) - \bar{f}(x_{k+1}))| && \text{(rearrangement)} \\ &\leq |f(x_k) - \bar{f}(x_k)| + |f(x_{k+1}) - \bar{f}(x_{k+1})| && \text{(triangle inequality)} \\ &\leq \varepsilon_f \max(1, |f(x_k)|) + \varepsilon_f \max(1, |f(x_{k+1})|) && \text{(eq. (2))} \\ &\leq 2\varepsilon_f \max(1, |f(x_k)|, |f(x_{k+1})|). && \text{(max property)} \end{aligned} \quad (16)$$

With these preparations, we now prove the lemma. From the condition $f(x_{k+1}) \leq f(x_k)$, eq. (16) can be further bounded as

$$\begin{aligned} 2\varepsilon_f \max(1, |f(x_k)|, |f(x_{k+1})|) &= 2\varepsilon_f \max(1, f(x_k), -f(x_{k+1})) && (f(x_{k+1}) \leq f(x_k)) \\ &\leq \frac{2\varepsilon_f}{1-\varepsilon_f} \max(1, \bar{f}(x_k), -\bar{f}(x_{k+1})) && \text{(eq. (14))} \\ &= \Delta_k, && \text{(eq. (4))} \end{aligned}$$

which yields the desired inequality with $\delta_k - \bar{\delta}_k \leq |\delta_k - \bar{\delta}_k| \leq \Delta_k$. $\square$

We also derive the following lemma for later use.

Lemma 2 *Under Assumption 4, it holds that*

$$\bar{\delta}_k \leq \delta_k + \frac{1+\varepsilon_f}{1-\varepsilon_f} \Delta_k.$$

Proof Since $-g_k^\top d_k \geq 0$ from Assumption 4 and eq. (9), the relaxed Armijo condition (eq. (4)) implies

$$\bar{f}(x_k) - \bar{f}(x_{k+1}) \geq -c\alpha_k g_k^\top d_k - \Delta_k \geq -\Delta_k. \quad (17)$$

Thus, using the general results in eq. (16) and $\Delta_k \geq 0$, we can further bound as

$$|\delta_k - \bar{\delta}_k| \leq 2\varepsilon_f \max(1, |f(x_k)|, |f(x_{k+1})|) \quad \text{(eq. (16))}$$

$$\leq \frac{2\varepsilon_f}{1-\varepsilon_f} \max(1, \bar{f}(x_k), -\bar{f}(x_k), \bar{f}(x_{k+1}), -\bar{f}(x_{k+1})) \quad (\text{eq. (14)})$$

$$\leq \frac{2\varepsilon_f}{1-\varepsilon_f} \max(1, \bar{f}(x_k) + \Delta_k, -\bar{f}(x_{k+1}) + \Delta_k) \quad (\text{eq. (17)})$$

$$\begin{aligned} &\leq \frac{2\varepsilon_f}{1-\varepsilon_f} (\max(1, \bar{f}(x_k), -\bar{f}(x_{k+1})) + \Delta_k) \\ &= \left(1 + \frac{2\varepsilon_f}{1-\varepsilon_f}\right) \Delta_k = \frac{1+\varepsilon_f}{1-\varepsilon_f} \Delta_k, \end{aligned} \quad (\text{eq. (4)})$$

which yields the desired inequality with $\bar{\delta}_k - \delta_k \leq \left| \delta_k - \bar{\delta}_k \right| \leq \frac{1+\varepsilon_f}{1-\varepsilon_f} \Delta_k$. $\square$

4.3 Backtracking Stage

We next analyze the line search in Algorithm 2. Recall that the parameter c in Algorithm 2 satisfies $0 < c < 1$ and $0 < m$ from Assumption 4.

Lemma 3 *Under Assumptions 3 and 4, the backtracking procedure in Algorithm 2 terminates whenever the step size α_k satisfies*

$$\alpha_k \leq \frac{2(1-c)m}{L}. \quad (18)$$

Proof Under the L -smoothness of f (Assumption 3), we always have

$$\begin{aligned} f(x_k) - f(x_k + \alpha_k d_k) &\geq -\alpha_k g_k^\top d_k - \frac{L}{2} \alpha_k^2 \|d_k\|^2 \\ &= -c\alpha_k g_k^\top d_k + \alpha_k \left(-(1-c)g_k^\top d_k - \frac{L\alpha_k}{2} \|d_k\|^2 \right). \end{aligned} \quad (19)$$

From Assumption 4 and eq. (9), we have $-g_k^\top d_k \geq m\|d_k\|^2$. Under eq. (18), we have

$$-(1-c)g_k^\top d_k - \frac{L\alpha_k}{2} \|d_k\|^2 \geq \left((1-c)m - \frac{L\alpha_k}{2} \right) \|d_k\|^2 \geq 0. \quad (20)$$

Combining eqs. (19) and (20) yields

$$f(x_k) - f(x_k + \alpha_k d_k) \geq -c\alpha_k g_k^\top d_k.$$

Since $-g_k^\top d_k \geq 0$ from Assumption 4 and eq. (9), the condition of Lemma 1 is satisfied with $x_{k+1} = x_k + \alpha_k d_k$. Thus, we obtain $\delta_k \leq \bar{\delta}_k + \Delta_k$, i.e.,

$$-c\alpha_k g_k^\top d_k \leq f(x_k) - f(x_k + \alpha_k d_k) \leq \bar{f}(x_k) - \bar{f}(x_k + \alpha_k d_k) + \Delta_k.$$

This inequality means that the relaxed Armijo condition (eq. (4)) is satisfied and thus the backtracking procedure terminates. This completes the proof. $\square$

Now, let us analyze Algorithm 2 using Lemma 3. Define

$$\bar{\alpha} := \beta_{\min} \min \left(\frac{2(1-c)m}{L}, 1 \right) > 0.$$

The following states the key property of the backtracking procedure.

Proposition 1 *Under Assumptions 3 and 4, the backtracking line search in Algorithm 2 terminates after finitely many rejections, and the step size α_k satisfies*

$$\alpha_k \geq \bar{\alpha}.$$

Proof Line 4 of Algorithm 2 multiplies α_k by β at each rejection, where $0 < \beta_{\min} \leq \beta \leq \beta_{\max} < 1$. For the number of backtracking rejections r , we have $\alpha_k \leq \beta_{\max}^r$. Thus, r is finite since the backtracking procedure terminates for sufficiently small α_k by Lemma 3. If $r = 0$, $\alpha_k = 1 \geq \bar{\alpha}$. Otherwise, $\alpha_k \geq \beta_{\min} \alpha'_k$, where α'_k is the last rejected step size. By Lemma 3, we have $\alpha'_k > \frac{2(1-c)m}{L}$, and thus

$$\alpha_k \geq \beta_{\min} \alpha'_k > \beta_{\min} \frac{2(1-c)m}{L} \geq \bar{\alpha}.$$

This completes the proof. $\square$

4.4 Bound for the case $\mu_k = 0$

In this subsection, we lower bound the sum of actual function decreases δ_k over $k \in K^0$. Recall that K^0 denotes the set of iteration indices k such that $\mu_k = 0$, as in eq. (5). To this end, we establish that the sum of Δ_k over $k \in K^0$ is upper bounded, where Δ_k is the error-absorbing term in the relaxed Armijo condition (eq. (4)). We introduce the following constant $\Delta_{\max}$, which serves as an upper bound for Δ_k of $k \in K^0$.

$$\Delta_{\max} := \frac{2\varepsilon_f}{1 - \varepsilon_f} \max(1, \bar{f}(x_0), -\bar{f}^*).$$

Lemma 4 *Under Assumptions 2 to 4, the sum of Δ_k over $k \in K^0$ satisfies*

$$\sum_{k \in K^0} \Delta_k \leq \bar{f}(x_0) - \bar{f}^* + \Delta_{\max}.$$

Proof Let $(k_i)_{i=0}^\ell$ with $\ell \geq 0$ be the elements of K^0 in increasing order. We have $\bar{f}(x_{k_\ell}) \leq \bar{f}(x_0)$ from eq. (6) with $k = k_\ell$, where the case $\ell = 0$ follows from $k_0 = 0$. Combining this with Assumption 2, we have $\Delta_{k_\ell} \leq \Delta_{\max}$. For all $i < \ell$, eq. (6) with $j = k_i$ and $k = k_{i+1}$ yields

$$\Delta_{k_i} \leq \bar{f}(x_{k_i}) - \bar{f}(x_{k_{i+1}}), \quad (21)$$

and summing Δ_k over $k \in K^0$ gives

$$\sum_{k \in K^0} \Delta_k \leq \sum_{i=0}^{\ell-1} (\bar{f}(x_{k_i}) - \bar{f}(x_{k_{i+1}})) + \Delta_{k_\ell} \quad (\text{eq. (21)})$$

$$\begin{aligned}
&= \bar{f}(x_0) - \bar{f}(x_{k_\ell}) + \Delta_{k_\ell} && \text{(telescoping sum)} \\
&\leq \bar{f}(x_0) - \bar{f}^* + \Delta_{\max}. && (-\bar{f}(x_{k_\ell}) \leq -\bar{f}^* \text{ and } \Delta_{k_\ell} \leq \Delta_{\max})
\end{aligned}$$

This completes the proof. $\square$

We now bound the sum of $\delta_k = f(x_k) - f(x_{k+1})$ over $k \in K^0$ with $\|g_k\|^2$ as follows.

Lemma 5 *Under Assumptions 2 to 4, the sum of δ_k over $k \in K^0$ satisfies*

$$\sum_{k \in K^0} \delta_k \geq \frac{c\bar{\alpha}}{M} \sum_{k \in K^0} \|g_k\|^2 - \frac{2}{1 - \varepsilon_f} \left(\bar{f}(x_0) - \bar{f}^* + \Delta_{\max} \right).$$

Proof The relaxed Armijo condition (eq. (4)) at iteration $k \in K^0$ yields

$$\begin{aligned}
\bar{\delta}_k + \Delta_k &= \bar{f}(x_k) - \bar{f}(x_{k+1}) + \Delta_k && \text{(eq. (13))} \\
&\geq -c\alpha_k g_k^\top d_k && \text{(eq. (4))} \\
&\geq \frac{c\bar{\alpha}}{M + \mu_k} \|g_k\|^2 && \text{(Proposition 1 and eq. (10))} \\
&= \frac{c\bar{\alpha}}{M} \|g_k\|^2. && (\mu_k = 0 \text{ since } k \in K^0) \tag{22}
\end{aligned}$$

Using Lemma 2, we also have

$$\bar{\delta}_k + \Delta_k \leq \delta_k + \frac{1 + \varepsilon_f}{1 - \varepsilon_f} \Delta_k + \Delta_k = \delta_k + \frac{2}{1 - \varepsilon_f} \Delta_k. \tag{23}$$

Then, combining eqs. (22) and (23) and summing over $k \in K^0$ gives

$$\sum_{k \in K^0} \frac{c\bar{\alpha}}{M} \|g_k\|^2 \leq \sum_{k \in K^0} \delta_k + \frac{2}{1 - \varepsilon_f} \sum_{k \in K^0} \Delta_k \leq \sum_{k \in K^0} \delta_k + \frac{2}{1 - \varepsilon_f} \left(\bar{f}(x_0) - \bar{f}^* + \Delta_{\max} \right),$$

where the last inequality follows from Lemma 4. Rearranging this completes the proof. $\square$

4.5 Bound for the case $\mu_k > 0$

Next, we show that the sum of δ_k over $k \in K^+$ is lower bounded. Recall that K^+ is the set of iteration indices k such that $\mu_k > 0$ as defined in eq. (5), and that μ_k is defined as $\theta_k \sqrt{\varsigma + \sum_{j \in K^+, j \leq k} \|g_j\|^2}$ with $\theta_k \in [\theta_{\min}, \theta_{\max}]$ in line 8 of Algorithm 1.

Before proceeding to the main proof, we present Lemma 6, which collects standard inequalities from the AdaGrad-Norm analysis [43, Lemma 3.2] [17, Lemma 3.1]. Its proof is deferred to appendix A.

Lemma 6 In Algorithm 1, we have

$$\begin{aligned} \sum_{k \in K^+} \frac{\|g_k\|^2}{\mu_k^2} &\leq \frac{1}{\theta_{\min}^2} \left(\ln \left(\varsigma + \sum_{k \in K^+} \|g_k\|^2 \right) - \ln(\varsigma) \right), \\ \sum_{k \in K^+} \frac{\|g_k\|^2}{\mu_k} &\geq \frac{1}{\theta_{\max}} \left(\sqrt{\varsigma + \sum_{k \in K^+} \|g_k\|^2} - \sqrt{\varsigma} \right). \end{aligned}$$

We now lower bound the sum of δ_k over $k \in K^+$ with $\|g_k\|^2$. We define the following constants for later use:

$$C_1 := \frac{\bar{\alpha}}{\theta_{\max}}, \quad C_2 := \frac{M\bar{\alpha} + L/2}{\theta_{\min}^2}.$$

Lemma 7 Under Assumptions 3 and 4, the sum of δ_k over $k \in K^+$ satisfies

$$\sum_{k \in K^+} \delta_k \geq C_1 \left(\sqrt{\varsigma + \sum_{k \in K^+} \|g_k\|^2} - \sqrt{\varsigma} \right) - C_2 \left(\ln \left(\varsigma + \sum_{k \in K^+} \|g_k\|^2 \right) - \ln(\varsigma) \right).$$

Proof From the L -smoothness (Assumption 3), for $k \in K^+$ we have

$$\begin{aligned} \delta_k &= f(x_k) - f(x_k + \alpha_k d_k) && \text{(eq. (12) and } x_{k+1} = x_k + \alpha_k d_k) \\ &\geq -\alpha_k g_k^\top d_k - \frac{\alpha_k^2 L}{2} \|d_k\|^2 && \text{(Assumption 3)} \\ &\geq \left(\frac{\alpha_k}{M + \mu_k} - \frac{\alpha_k^2 L}{2(m + \mu_k)^2} \right) \|g_k\|^2 && \text{(eqs. (10) and (11))} \\ &\geq \left(\frac{\bar{\alpha}}{M + \mu_k} - \frac{L}{2(m + \mu_k)^2} \right) \|g_k\|^2. && \text{(Proposition 1 and } \alpha_k \leq 1) \end{aligned} \quad (24)$$

We can further bound the terms in eq. (24) as

$$\frac{\bar{\alpha}}{M + \mu_k} = \frac{\bar{\alpha}}{\mu_k} \frac{1}{1 + \frac{M}{\mu_k}} \geq \frac{\bar{\alpha}}{\mu_k} \left(1 - \frac{M}{\mu_k} \right) = \frac{\bar{\alpha}}{\mu_k} - \frac{M\bar{\alpha}}{\mu_k^2}, \quad \frac{L}{2(m + \mu_k)^2} \leq \frac{L}{2\mu_k^2}. \quad (25)$$

Thus, applying eq. (25) to eq. (24) yields

$$\delta_k \geq \left(\frac{\bar{\alpha}}{\mu_k} - \frac{M\bar{\alpha} + L/2}{\mu_k^2} \right) \|g_k\|^2. \quad (26)$$

Summing eq. (26) over $k \in K^+$ and applying Lemma 6 yield

$$\begin{aligned} \sum_{k \in K^+} \delta_k &\geq \sum_{k \in K^+} \left(\frac{\bar{\alpha}}{\mu_k} - \frac{M\bar{\alpha} + L/2}{\mu_k^2} \right) \|g_k\|^2 \\ &\geq \frac{\bar{\alpha}}{\theta_{\max}} \left(\sqrt{\varsigma + \sum_{k \in K^+} \|g_k\|^2} - \sqrt{\varsigma} \right) - \frac{M\bar{\alpha} + L/2}{\theta_{\min}^2} \left(\ln \left(\varsigma + \sum_{k \in K^+} \|g_k\|^2 \right) - \ln(\varsigma) \right), \end{aligned}$$

which yields the desired result. $\square$

4.6 Global Convergence Rate

Finally, we combine the results for $k \in K^0$ stated in Lemma 5 and $k \in K^+$ stated in Lemma 7 to derive an upper bound on the total sum of $\|g_k\|^2$. We define a constant F as follows:

$$F := f(x_0) - f^* + \frac{2}{1 - \varepsilon_f} \left(\bar{f}(x_0) - \bar{f}^* + \Delta_{\max} \right) + C_1 \sqrt{\varsigma} - C_2 \ln(\varsigma). \quad (27)$$

Then, we can obtain the following unified bound for the two types of iterations.

Lemma 8 *Under Assumptions 2 to 4, the sum of $\|g_k\|^2$ over $k \in K = K^0 \cup K^+$ satisfies*

$$F \geq \frac{c\bar{\alpha}}{M} \sum_{k \in K^0} \|g_k\|^2 + C_1 \sqrt{\varsigma + \sum_{k \in K^+} \|g_k\|^2} - C_2 \ln \left(\varsigma + \sum_{k \in K^+} \|g_k\|^2 \right),$$

Proof Since $f(x_{k_{\max}}) \geq f^*$ by Assumption 2 and $K = K^0 \cup K^+$, we have

$$f(x_0) - f^* \geq f(x_0) - f(x_{k_{\max}}) = \sum_{k \in K} \delta_k = \sum_{k \in K^0} \delta_k + \sum_{k \in K^+} \delta_k.$$

From Lemmas 5 and 7, we can further bound as

$$\begin{aligned} f(x_0) - f^* &\geq \frac{c\bar{\alpha}}{M} \sum_{k \in K^0} \|g_k\|^2 - \frac{2}{1 - \varepsilon_f} \left(\bar{f}(x_0) - \bar{f}^* + \Delta_{\max} \right) \\ &\quad + C_1 \sqrt{\varsigma + \sum_{k \in K^+} \|g_k\|^2} - C_1 \sqrt{\varsigma} - C_2 \ln \left(\varsigma + \sum_{k \in K^+} \|g_k\|^2 \right) + C_2 \ln(\varsigma). \end{aligned}$$

Substituting eq. (27) yields the desired bound, concluding the proof. $\square$

Now, we establish the following convergence rate for Algorithm 1.

Theorem 1 *Under Assumptions 1 to 4, the average of squared gradient norms generated by Algorithm 1 satisfies*

$$\frac{1}{k_{\max}} \sum_{k=0}^{k_{\max}-1} \|g_k\|^2 = \mathcal{O} \left(\frac{1}{k_{\max}} \right).$$

Proof Let us define the function $\phi: \mathbb{R}_{>0} \rightarrow \mathbb{R}$ as

$$\phi(G) := \frac{C_1}{2} \sqrt{G} - C_2 \ln(G). \quad (28)$$

Since $\phi'(G) = \frac{C_1}{4\sqrt{G}} - \frac{C_2}{G}$, its minimum over $G > 0$ is attained at $G^* := \left(\frac{4C_2}{C_1} \right)^2$ with

$$\phi(G^*) = 2C_2 \left(1 - \ln \left(\frac{4C_2}{C_1} \right) \right) = 2C_2 \ln \left(\frac{eC_1}{4C_2} \right),$$

where e is the base of the natural logarithm. Thus, Lemma 8 yields

$$\begin{aligned} F &\geq \frac{c\bar{\alpha}}{M} \sum_{k \in K^0} \|g_k\|^2 + \frac{C_1}{2} \sqrt{\sum_{k \in K^+} \|g_k\|^2} + \phi\left(\sum_{k \in K^+} \|g_k\|^2 + \varsigma\right) \\ &\geq \frac{c\bar{\alpha}}{M} \sum_{k \in K^0} \|g_k\|^2 + \frac{C_1}{2} \sqrt{\sum_{k \in K^+} \|g_k\|^2} + \phi(G^*), \end{aligned}$$

which is equivalent to

$$0 \leq \frac{c\bar{\alpha}}{M} \sum_{k \in K^0} \|g_k\|^2 + \frac{C_1}{2} \sqrt{\sum_{k \in K^+} \|g_k\|^2} \leq F' \quad (29)$$

where

$$\begin{aligned} F' &:= F - \phi(G^*) \\ &= f(x_0) - f^* + \frac{2}{1 - \varepsilon_f} \left(\bar{f}(x_0) - \bar{f}^* + \Delta_{\max} \right) + C_1 \sqrt{\varsigma} - C_2 \ln(\varsigma) - 2C_2 \ln\left(\frac{eC_1}{4C_2}\right). \end{aligned}$$

It remains to bound $\sum_{k \in K} \|g_k\|^2 = \sum_{k \in K^0} \|g_k\|^2 + \sum_{k \in K^+} \|g_k\|^2$ given the constraint in eq. (29). More generally, let $a, b > 0$, $c \geq 0$, and $x, y \geq 0$, and consider maximizing $x + y$ subject to $ax + b\sqrt{y} \leq c$. The maximum is achieved on the boundary $ax + b\sqrt{y} = c$ since $x + y$ is increasing in both variables. On this boundary, $y = ((c - ax)/b)^2$, and thus $x + y$ is convex on $0 \leq x \leq c/a$, attaining its maximum at an endpoint. Evaluating the endpoints yields $x + y \leq \max\{c/a, (c/b)^2\}$. Applying this result to eq. (29) gives

$$\begin{aligned} \frac{1}{k_{\max}} \sum_{k=0}^{k_{\max}-1} \|g_k\|^2 &= \frac{1}{k_{\max}} \left(\sum_{k \in K^0} \|g_k\|^2 + \sum_{k \in K^+} \|g_k\|^2 \right) \quad (K = K^0 \cup K^+) \\ &\leq \frac{1}{k_{\max}} \max\left(\frac{MF'}{c\bar{\alpha}}, \frac{4F'^2}{C_1^2}\right), \end{aligned} \quad (30)$$

which completes the proof. $\square$

As a remark, the function $\phi(G)$ in eq. (28) is closely related to the Lambert W function [8, 17]. For $x \in (-1/e, 0)$, the equation $ye^y = x$ admits two real negative solutions, and the smaller one is given by the second branch of the Lambert W function, $y = W_{-1}(x) < 0$. Using this function, the equation $\phi(G) = 0$ can be solved in closed form, which leads to a sharper bound in Theorem 1. Since this is not essential for the main argument, we omit the details and refer to [17].

We next interpret eq. (30) in the proof of Theorem 1. The contribution of the indices in K^0 is analogous to the classical analysis of gradient descent. Indeed, for gradient descent with fixed step size $\alpha = 1/L$, one has

$$\frac{1}{k_{\max}} \sum_{k=0}^{k_{\max}-1} \|g_k\|^2 \leq \frac{1}{k_{\max}} \frac{2(f(x_0) - f^*)}{\alpha}.$$

Thus, the term associated with K^0 reflects the standard $\mathcal{O}(f(x_0) - f^*)$ dependence of first-order methods since F' contains the initial gaps $f(x_0) - f^*$ and $\bar{f}(x_0) - \bar{f}^*$. In contrast, the contribution of K^+ resembles the behavior of AdaGrad-Norm, whose convergence bound scales quadratically with the initial optimality gap [43].

Finally, Theorem 1 implies the iteration-complexity bound

$$\min \{k \mid \|\nabla f(x_k)\| \leq \varepsilon_{\text{gtol}}\} = \mathcal{O}(1/\varepsilon_{\text{gtol}}^2).$$

More explicitly, eq. (30) gives the sufficient bound

$$k_{\max} \geq \frac{1}{\varepsilon_{\text{gtol}}^2} \max \left\{ \frac{MF'}{c\bar{\alpha}}, \frac{4(F')^2}{C_1^2} \right\}.$$

For fixed algorithmic parameters, define $A := \max\{1, L/m\}$. Then,

$$\bar{\alpha}^{-1} = \mathcal{O}(A), \quad C_1^{-1} = \mathcal{O}(A), \quad C_2 = \mathcal{O}(M/A + L).$$

Substituting these bounds and the definition of $\Delta_{\max}$ into F' gives

$$F' = \mathcal{O} \left(f(x_0) - f^* + \frac{\bar{f}(x_0) - \bar{f}^*}{1 - \varepsilon_f} + \frac{\varepsilon_f \max\{1, \bar{f}(x_0), -\bar{f}^*\}}{(1 - \varepsilon_f)^2} + 1 + (M/A + L) \ln(2 + M + LA) \right).$$

Consequently, the sufficient iteration bound is $\mathcal{O}(\varepsilon_{\text{gtol}}^{-2} \max\{MAF', A^2(F')^2\})$. For the construction in section 5.1, fixed algorithmic parameters give fixed $m > 0$ and $M = \mathcal{O}(\max\{1, \|g_0\|\})$. Thus, m , M , and F' are independent of the target accuracy $\varepsilon_{\text{gtol}}$, while ε_f , the initial gaps, and the smoothness and spectral bounds affect the prefactor.

5 Practical Choices of Algorithmic Variables

In sections 3 and 4, we introduced Algorithm 1 and established its convergence properties. In this section, we discuss practical choices of algorithmic variables and parameters that lead to fast and stable performance. The quantities to be discussed here are the matrices B_k , the step sizes α_k , and the regularization parameters μ_k .

5.1 Approximate Hessian B_k

We first discuss the construction of the matrices B_k used in line 9 of Algorithm 1. We show that the shifted L-BFGS construction with the curvature-pair selection rule below satisfies Assumption 4.

Construction of B_k : We recall the L-BFGS construction [3, 25, 31]. For an iteration k , we define

$$s_k := x_{k+1} - x_k, \quad y_k := g_{k+1} - g_k$$

where g_k and g_{k+1} are the gradients at x_k and x_{k+1} , respectively. Starting from an initial matrix, an approximate Hessian is obtained by successively applying BFGS updates to a sequence of curvature pairs. Let $\{(s_{k_i}, y_{k_i})\}_{i=0}^{p-1}$ be a collection of curvature pairs ordered so that $k_0 < \dots < k_{p-1}$. Starting from the initial matrix $B^{(0)} = \gamma I$ with $\gamma = \|y_{k_{p-1}}\|^2 / y_{k_{p-1}}^\top s_{k_{p-1}}$, we define $\text{BFGS}(\{(s_{k_i}, y_{k_i})\}_{i=0}^{p-1})$ as the matrix $B^{(p)}$ obtained by sequentially applying the BFGS update as

$$B^{(j+1)} = B^{(j)} - \frac{B^{(j)} s_{k_j} s_{k_j}^\top B^{(j)}}{s_{k_j}^\top B^{(j)} s_{k_j}} + \frac{y_{k_j} y_{k_j}^\top}{y_{k_j}^\top s_{k_j}}$$

for $j = 0, 1, \dots, p-1$, where each curvature pair (s_{k_j}, y_{k_j}) is applied in order. Here, $p > 0$ denotes the maximum number of stored curvature pairs and is a fixed small integer in the L-BFGS method. When fewer than p curvature pairs are available, the BFGS update is applied using all currently available pairs.

We now give a sufficient condition for the upper spectral bound in Assumption 4, based on variants of existing results [13, Theorem 5.5] [15, Lemma 1]. Its proof is deferred to appendix B.

Proposition 2 *Let $\{(s_{k_i}, y_{k_i})\}_{i=0}^{p-1}$ be curvature pairs with $s_{k_i} \neq 0$. Suppose that there exists a constant $\Lambda > 0$ such that every pair (s, y) in this collection satisfies*

$$y^\top s \geq \frac{1}{\Lambda} \|y\|^2 > 0. \quad (31)$$

Then

$$0 \prec \text{BFGS}(\{(s_{k_i}, y_{k_i})\}_{i=0}^{p-1}) \preceq (1+p)\Lambda I. \quad (32)$$

Proposition 2 supplies the upper bound, while a fixed diagonal shift will supply the lower bound in Assumption 4. We first apply a scalar Powell-type damping to a candidate pair with $y_{k_i}^\top s_{k_i} < 0$. For a parameter $\eta_{\text{damp}} \in (0, 1)$, let $b_i = \|y_{k_i}\|^2 / |y_{k_i}^\top s_{k_i}|$ and set

$$\bar{y}_{k_i} = \theta_i^{\text{damp}} y_{k_i} + (1 - \theta_i^{\text{damp}}) b_i s_{k_i}, \quad \theta_i^{\text{damp}} = \frac{\eta_{\text{damp}} b_i \|s_{k_i}\|^2}{b_i \|s_{k_i}\|^2 - y_{k_i}^\top s_{k_i}} \in [0, 1).$$

If $y_{k_i}^\top s_{k_i} \geq 0$, we set $\bar{y}_{k_i} = y_{k_i}$. Candidate pairs proceed to cautious selection only when the damping can be computed safely.

We then select the damped pairs using a cautious update in the spirit of [23, 26, 27]. Define the fixed initial scale and diagonal shift by

$$\rho_0 := \max\{1, \|g_0\|\}, \quad \omega_{\min} := \min\{1, \sqrt{\zeta}\} > 0.$$

For a fixed $\Lambda \geq 1$, a candidate pair $(s, \bar{y})$ is retained only if

$$\bar{y}^\top s \geq \frac{\|\bar{y}\|^2}{\Lambda \rho_0} > 0. \quad (33)$$

Consequently, every retained pair $(s, \bar{y}/\rho_0)$ satisfies eq. (31), and the homogeneity of the BFGS update gives

$$\text{BFGS}(\{(s_{k_i}, \bar{y}_{k_i})\}) = \rho_0 \text{BFGS}(\{(s_{k_i}, \bar{y}_{k_i}/\rho_0)\}).$$

Using at most p retained pairs, we define

$$B_k = \begin{cases} \omega_{\min} I + \text{BFGS}(\{(s_{k_i}, \bar{y}_{k_i})\}), & \text{if the memory is nonempty,} \\ (\omega_{\min} + \rho_0) I, & \text{otherwise.} \end{cases} \quad (34)$$

It follows from Proposition 2, the homogeneity identity, and $\omega_{\min} \leq 1 \leq \rho_0$ that

$$\omega_{\min} I \preceq B_k \preceq (1 + (1 + p)\Lambda) \rho_0 I.$$

Thus, Assumption 4 holds with the iteration-independent constants $m = \omega_{\min}$ and $M = (1 + (1 + p)\Lambda) \rho_0$. This argument uses neither convexity nor cocoercivity of ∇f . In our implementation, we use $p = 10$, $\Lambda = 10^8$, and $\eta_{\text{damp}} = 0.8$, and strengthen eq. (33) with the absolute safeguard $\bar{y}^\top s \geq 10^{-16}$ to reject numerically degenerate pairs.

Exact Compact Solve: We use a compact representation of the raw L-BFGS matrix to solve the regularized system exactly [22]. Specifically, we compute $d_k = -(B_k + \mu_k I)^{-1} g_k$ for the matrix in eq. (34), without absorbing the additive regularization into the curvature pairs. This exact compact solve therefore coincides with the direction analyzed in section 4. When the memory is empty, the same definition reduces to $d_k = -g_k/(\rho_0 + \omega_{\min} + \mu_k)$. As a numerical safeguard when the compact factorization is unavailable or does not return a finite descent direction, the implementation uses this zero-memory direction, which also corresponds to a matrix satisfying Assumption 4.

Shifted-Pair Approximation: For comparison, the implementation retains the simpler shifted-pair approximation

$$B_k + \mu_k I \approx \text{BFGS}(\{(s_{k_i}, \bar{y}_{k_i} + (\mu_k + \omega_{\min}) s_{k_i})_{i=0}^{p-1}\}). \quad (35)$$

Only Ours-AR uses this two-loop approximation, together with the optional restart in section 5.3. The analysis applies to the exact compact solve used by Ours.

Modified Secant Correction: Both proposed variants apply a function-based modified secant correction before the damping and cautious selection described above [45, 47]. A scale-aware safeguard suppresses the correction whenever it would dominate the raw gradient difference. The final corrected and damped pair must still satisfy eq. (33), so this modification preserves the spectral guarantee for Ours. The correction formula and safeguard are specified in our open-source implementation.

5.2 Step Size α_k

We next discuss the adjustment of the step size α_k in line 4 of Algorithm 2. As indicated in the pseudo-code, we choose α_k using interpolation of the objective function. Moré–Thuente line searches [28] are widely used in quasi-Newton methods [41], and they typically employ cubic or quadratic interpolation to select trial step sizes. In our implementation, we use an essentially identical procedure, except that the trial step size is obtained by clipping it to the interval $[\beta_{\min}\alpha_k, \beta_{\max}\alpha_k]$. For all numerical experiments, we set the Armijo parameter to $c = 10^{-4}$ and choose the interpolation bounds as $\beta_{\min} = 1/16$ and $\beta_{\max} = 15/16$.

Furthermore, when $\mu_k > 0$ and $\alpha_k = 1$, we introduce a one-time adjustment of the trial step size using gradient information. Let d_k be the search direction and g_k and g_{try} the gradients at the current and trial points. If the directional derivative along d_k changes sign, $d_k^\top g_k < 0$ and $d_k^\top g_{\text{try}} > 0$, and the trial gradient is sufficiently aligned with d_k , namely $d_k^\top g_{\text{try}} > 0.5\|d_k\|\|g_{\text{try}}\|$, we rescale α_k once by a secant-like factor,

$$\alpha_k \leftarrow \text{clip}\left(\alpha_k \frac{-d_k^\top g_k}{d_k^\top g_{\text{try}} - d_k^\top g_k}, \beta_{\min}\alpha_k, \beta_{\max}\alpha_k\right),$$

where $\text{clip}(x, a, b) := \min(\max(x, a), b)$ is a clipping function. When objective values are unreliable and backtracking and interpolation do not occur, this correction effectively refines the step size. After the adjustment, the relaxed Armijo condition is evaluated again, and ordinary backtracking continues if necessary. The clipping interval gives $\alpha_k \geq \beta_{\min}$ whenever this adjustment is applied, so the conservative lower bound $\bar{\alpha}$ in Proposition 1 and the subsequent convergence analysis remain valid.

5.3 Regularization Parameter μ_k

We finally discuss the adjustment of the regularization parameter μ_k in line 8 of Algorithm 1.

Regularization Update: The OFFO accumulator contains exactly the gradient norms generated during regularized iterations:

$$G_k^2 = \varsigma + \sum_{j \in K^+, j \leq k} \|g_j\|^2.$$

For a parameter $\eta_\mu > 0$, we set

$$\mu_k = \text{clip}\left(\eta_\mu \|g_k\|, \theta_{\min} G_k, \theta_{\max} G_k\right).$$

This clipping rule gives $\mu_k = \theta_k G_k$ for some $\theta_k \in [\theta_{\min}, \theta_{\max}]$, as required by eq. (7). In our implementation, $\eta_\mu = 1/10$, $\theta_{\min} = 1/100$, $\theta_{\max} = 1$, and $\varsigma = 10^{-20}$, so the initial accumulator scale is $G_{-1} = \sqrt{\varsigma} = 10^{-10}$.

Optional Accumulator Restart: The optional restart is a practical heuristic intended to reduce over-regularization after clear progress in the objective value. For a fixed threshold $\tau > 0$, the heuristic resets only the accumulator when the method returns from K^+ to K^0 and

$$\min_{j \in K^0, j < k} (\bar{f}(x_j) - \Delta_j) - \bar{f}(x_k) \geq \tau. \quad (36)$$

The sets K^0 and K^+ and the reference values in eq. (36) are not reset. The numerical experiments distinguish the no-restart and restarted variants. The former coincides with the analyzed accumulator, whereas the latter uses $\tau = 1$ and at most 10 restarts.

6 Numerical Experiments

We conducted numerical experiments to evaluate the practical performance of our proposed method and to compare it with existing optimization algorithms.

6.1 Methods

We compared the following methods in our experiments.

- (Ours): Our proposed regularized quasi-Newton method
 - We used Algorithms 1 and 2 with the details in section 5.
 - We used the modified secant condition described in section 5.1.
 - (Ours-AR) denotes the variant with shifted-pair approximation (A) and accumulator restart (R), as described in eq. (35) and section 5.3.
- (Line): Line search L-BFGS method
 - Our Python implementation ported from LBFGSpp [32].
 - The step size was determined using the Moré–Thuente line search, closely resembling SciPy’s implementation.
- (Reg): Regularized L-BFGS method [22]
 - A quadratic regularization-based quasi-Newton method that does not rely on line search and is conceptually closer to trust-region methods.
 - We wrapped the function `regLBFGS` with `solveNonmonotone` from [22].
 - (Reg-Sec) denotes the variant using the approximate computation in eq. (35). It wraps the `regLBFGSsec` function from [22].
- (SciPy): SciPy’s L-BFGS-B method [41]
 - An established L-BFGS method implemented in C and called from Python.
 - We used the default parameters, with the exception that `ftol` was set to 0 to avoid premature termination.
- (NTQN): Noise-tolerant quasi-Newton method [37, 44]
 - A method designed to accommodate inexact gradient information, as mentioned in section 2.3.
 - We used the common gradient-tolerance stopping rule rather than NTQN’s recommended termination mode, which yielded a substantially lower attainment rate under the common tolerance.
- (ASTR1-Adagrad): The purely first-order adagrad variant of ASTR1.

- The parameter setting is the same as reported by Gratton et al. [17, Table 1].

In the following, each method is referred to by the name given in parentheses. Except for the first-order ASTR1-Adagrad baseline, all these methods are L-BFGS-based, and we set the number of stored curvature pairs to 10, the default value in SciPy’s L-BFGS-B implementation. All experiments were conducted on an Apple Mac Pro (Late 2013) with a 2.7 GHz 12-core Intel Xeon E5-2697 v2 processor and 64 GB of memory. All computations were performed using the CPU only.

6.2 Experimental Results

6.2.1 Test Problems and Settings

We evaluated all algorithms on unconstrained problems from the CUTEst benchmark collection [14], implemented in Python using the PyCUTEst interface [12]. All problems were initialized with the default starting points provided by the library. Following previous work [22], we excluded problems whose initial points were already stationary or whose function values or gradients contained NaN or INF. **The retained problems have dimensions ranging from 2 to 123,200.**

We organize the CUTEst experiments into the following two groups according to where their results are reported:

- *Artificial-noise experiments* on 220 problems using 64-bit arithmetic (results in section 6.2.3)
 - **Function-value noise:** We add independent noise $\xi_f \sim \text{unif}(-10^{-3}, 10^{-3})$ to each function evaluation while retaining exact gradients. We set the nominal error allowance to $\epsilon_f = 10^{-2}$.
 - **Joint function-and-gradient noise:** We use the same function perturbation and additionally perturb each gradient component independently by $\xi_g^{(i)} \sim \text{unif}(-10^{-3}, 10^{-3})$. We again set $\epsilon_f = 10^{-2}$.
 - **Error-bound sensitivity:** Using the function-value-noise setting, we compare the underestimated allowance $\epsilon_f = 10^{-4}$, the nominal allowance $\epsilon_f = 10^{-2}$, and the overestimated allowance $\epsilon_f = 10^{-1}$.
- *Floating-point-precision experiments* (results in section 6.2.4)
 - **64-bit evaluations:** 220 problems with $\epsilon_f = 2.22 \times 10^{-9}$.
 - **32-bit evaluations:** 220 problems with $\epsilon_f = 1.19 \times 10^{-3}$.
 - **16-bit evaluations:** 152 problems with $\epsilon_f = 9.77 \times 10^{-2}$.

Here, $\text{unif}(a, b)$ denotes a uniform random variable in the interval $[a, b]$. The additive uniform-noise model follows the model used by Shi et al. [37]. Each artificial-noise setting uses the five seeds 0, 1, 2, 3, 4. For NTQN in the function-value-noise and joint-noise experiments, we supplied the conservative absolute noise bound $\text{eps}_f = 10^{-2}$ because each function perturbation is bounded in magnitude by 10^{-3} .

In the reduced precision settings, we first cast the input vector x to the target precision, evaluate CUTEst in 64-bit precision at the rounded input, and then round the returned function value and gradient to the target precision. The solver receives these rounded quantities as 64-bit values, so this tests reduced-precision input and

explanation	min abs value	machine epsilon	ϵ_f
64-bit double precision	2.23×10^{-308}	$2^{-52} \approx 2.22 \times 10^{-16}$	2.22×10^{-9}
32-bit single precision	1.18×10^{-38}	$2^{-23} \approx 1.19 \times 10^{-7}$	1.19×10^{-3}
16-bit half precision	6.10×10^{-5}	$2^{-10} \approx 9.77 \times 10^{-4}$	9.77×10^{-2}

Table 2: The “min abs value” is the smallest positive normalized number, “machine epsilon” is the spacing from 1 to the next larger representable number, and ϵ_f is the parameter in (2) used in our experiments.

output rather than entirely low-precision arithmetic. Since the returned gradient $g(x)$ need not equal $\nabla f(x)$, the reduced-precision and joint-noise experiments complement the exact-gradient analysis by evaluating performance beyond its assumptions.

The error allowances for the three precision settings are summarized in Table 2. These values are empirical safety margins for accumulated floating-point errors rather than certified bounds. The multipliers of machine epsilon are 10^7 , 10^4 , and 10^2 for 64-, 32-, and 16-bit precision, respectively. We reduce the multiplier at lower precision to retain $\epsilon_f < 1$ while allowing errors larger than a single rounding error. **Because the appropriate value of ϵ_f may not be known accurately in practice, the sensitivity experiment examines how under- and overestimation affect performance.** We also set the maximum number of iterations to $k_{\max} = 15,000$ and timed out runs exceeding 10 minutes per problem.

6.2.2 Performance Profiles

We used performance profiles [10] to compare the algorithms. Let P denote the set of test instances and S the set of solvers. For each instance $p \in P$ and solver $s \in S$, let $n_{p,s} > 0$ denote the number of oracle calls (i.e., function and gradient evaluations) required to reach a specified gradient norm tolerance ϵ_{gtol} . In this experiment, we use the number of oracle calls rather than clock time to avoid the influence of algorithmic overhead and to focus on the intrinsic optimization efficiency. When a solver does not reach the gradient tolerance before termination or the iteration or time limit, we set $n_{p,s} = +\infty$. Then, the performance ratio $r_{p,s}$ is defined as

$$r_{p,s} = \frac{n_{p,s}}{\min_{s' \in S} n_{p,s'}}.$$

If no solver succeeds on an instance, we set all its performance ratios to $+\infty$. For a given threshold $\tau \geq 1$ and solver s , the performance profile plots the proportion of test instances for which $r_{p,s} \leq \tau$. Thus, the y-coordinate for solver s at x-coordinate τ is given by

$$\rho_s(\tau) = \frac{1}{|P|} |\{p \in P \mid r_{p,s} \leq \tau\}|.$$

A curve that lies above others indicates better performance. Following standard implementations [41], we used the infinity norm for the gradient, though other norms may equally be employed. **All reported success percentages use the same common gradient tolerance as the corresponding profiles. A run that terminates without attaining that tolerance is counted as unsolved for this metric.**

Method	Function-value noise		Joint noise	
	Solved (%)	Calls	Solved (%)	Calls
Ours	87.0 ± 0.69	68	84.8 ± 0.69	71
Ours-AR	86.1 ± 0.52	65	83.6 ± 0.32	64
NTQN	81.2 ± 1.49	64	71.9 ± 0.38	62
Reg	77.8 ± 1.13	60	73.5 ± 1.35	56
Reg-Sec	74.0 ± 0.87	54	73.0 ± 1.35	62
ASTR1-Adagrad	54.5 ± 0.00	300	52.6 ± 0.20	295

Table 3: Success rates and conditional median oracle calls at $\varepsilon_{\text{gtol}} = 10^{-2}$ in the additive-noise experiments. Each setting contains 1,100 problem-seed instances. Success rates are reported as the mean $\pm$ sample standard deviation over the five seeds, in percentage points, and oracle-call medians are computed over successful runs.

6.2.3 Results with Artificial Noise

Table 3 summarizes the function-value noise experiment and the joint noise experiment described in section 6.2.1, and Figures 1 and 2 shows their performance profiles. The former satisfies the assumptions of section 4, whereas the latter tests empirical robustness beyond the exact-gradient theory.

Both proposed variants solve at least one instance of dimension 123,200, the largest dimension in the retained collection. In the theory-aligned function-value-noise setting, Ours and Ours-AR solve 87.0% and 86.1% of the instances, respectively, compared with 81.2% for NTQN. The joint-noise results provide empirical evidence beyond the exact-gradient setting, while the convergence result remains specific to exact gradients. Under NTQN’s recommended termination mode, 50.1% of the same function-noise instances attain the common tolerance, compared with 81.2% when the external common-tolerance stop is used. All recommended-mode runs terminate without timeout or exception, and 619 of them stop because NTQN detects that function-value changes have reached its estimated function-noise level (flag 4). The lower scored fraction therefore mainly reflects NTQN’s noise-dependent stopping rule rather than failed execution.

Figure 3 shows the performance profiles for the error-bound sensitivity experiment. The combined profiles compare both variants under all three supplied bounds. The nominal bound gives the highest solved percentage for Ours. Ours-AR is less sensitive to overestimation. Underestimation need not satisfy the assumed error model, and the results show no consistent performance benefit from it.

6.2.4 Results at Different Precision Levels

We next present performance profiles for each floating-point precision (64-, 32-, and 16-bit). We evaluated each method at three gradient norm tolerance levels, i.e., $\varepsilon_{\text{gtol}} \in \{10^{-1}, 10^{-3}, 10^{-5}\}$. These tolerance levels represent different optimization regimes, from moderate accuracy to high precision requirements. Figure 4 shows the performance profiles across all three precision levels. **At the strictest tolerance**

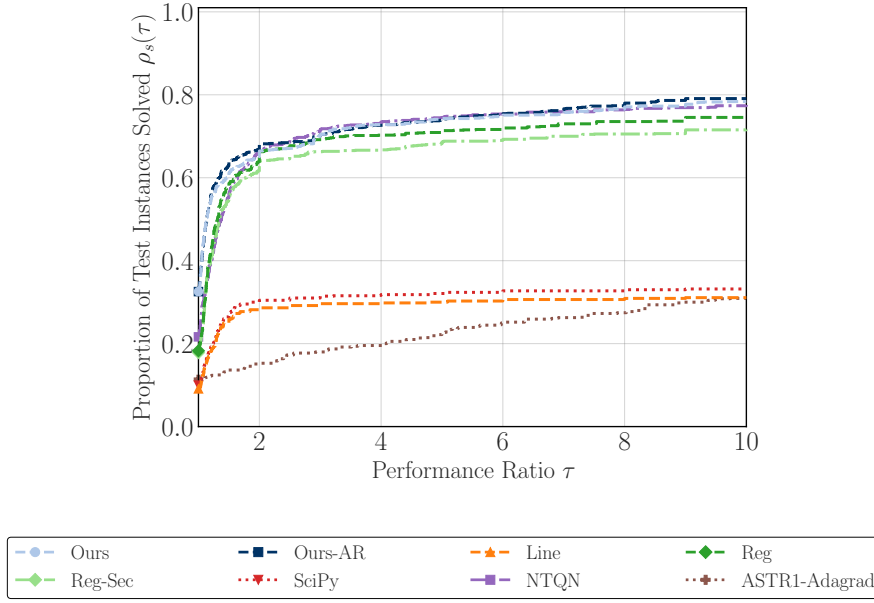


Fig. 1: Theory-aligned function-value-noise results at noise level 10^{-3} , supplied error bound $\epsilon_f = 10^{-2}$, and $\epsilon_{\text{gtol}} = 10^{-2}$.

$\epsilon_{\text{gtol}} = 10^{-5}$, Ours and Ours-AR solve 86.4% and 86.8% of the 64-bit problems, 52.3% and 53.6% of the 32-bit problems, and 28.9% and 34.9% of the 16-bit problems. The rates quantify how attained accuracy changes as oracle precision is reduced, while both variants retain higher solved percentages than all existing quasi-Newton baselines at 16- and 32-bit precision. The relative ranking of the methods varies with the tolerance and precision.

6.3 Computational Time

Here, runtime means the elapsed time until the first iterate attaining the common gradient tolerance. A run that does not attain the tolerance is counted as unsolved. To assess end-to-end computational cost together with reduced-precision stability, Figure 5 plots the fraction of 32-bit problems attaining $\epsilon_{\text{gtol}} = 10^{-3}$ directly against runtime. At this tolerance, Ours solves 74.5% of the problems and Ours-AR solves 73.6%, compared with 63.6% for the best existing baseline.

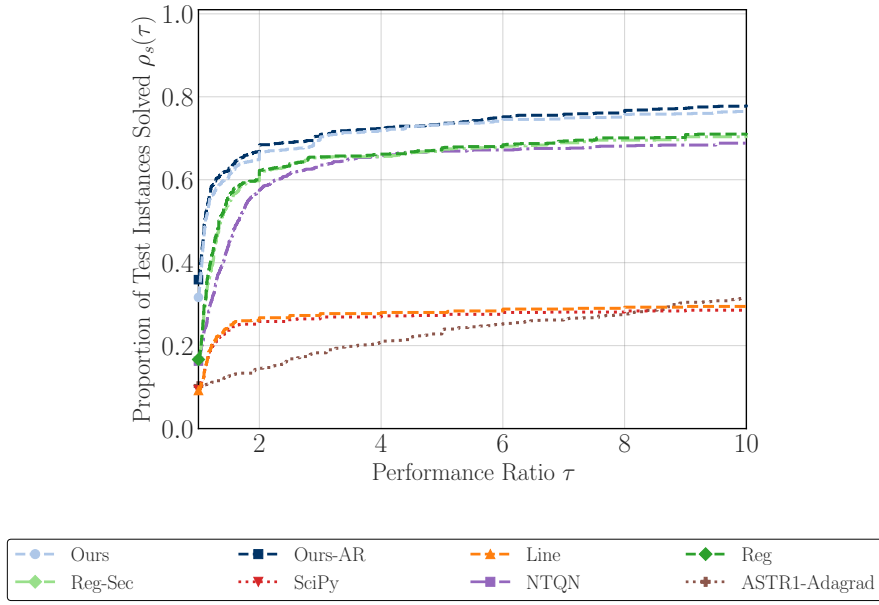


Fig. 2: Performance profile for the deliberately adverse joint function-and-gradient noise setting (componentwise level 10^{-3}) and $\epsilon_{\text{gtol}} = 10^{-2}$.

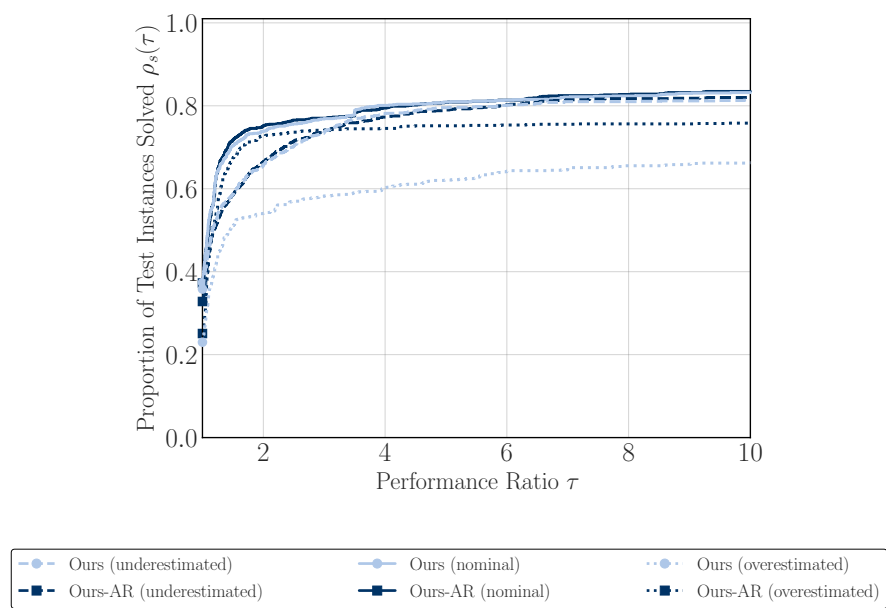


Fig. 3: Performance profiles for error-bound misspecification under function-value-only noise of level 10^{-3} and $\varepsilon_{\text{gtol}} = 10^{-2}$.

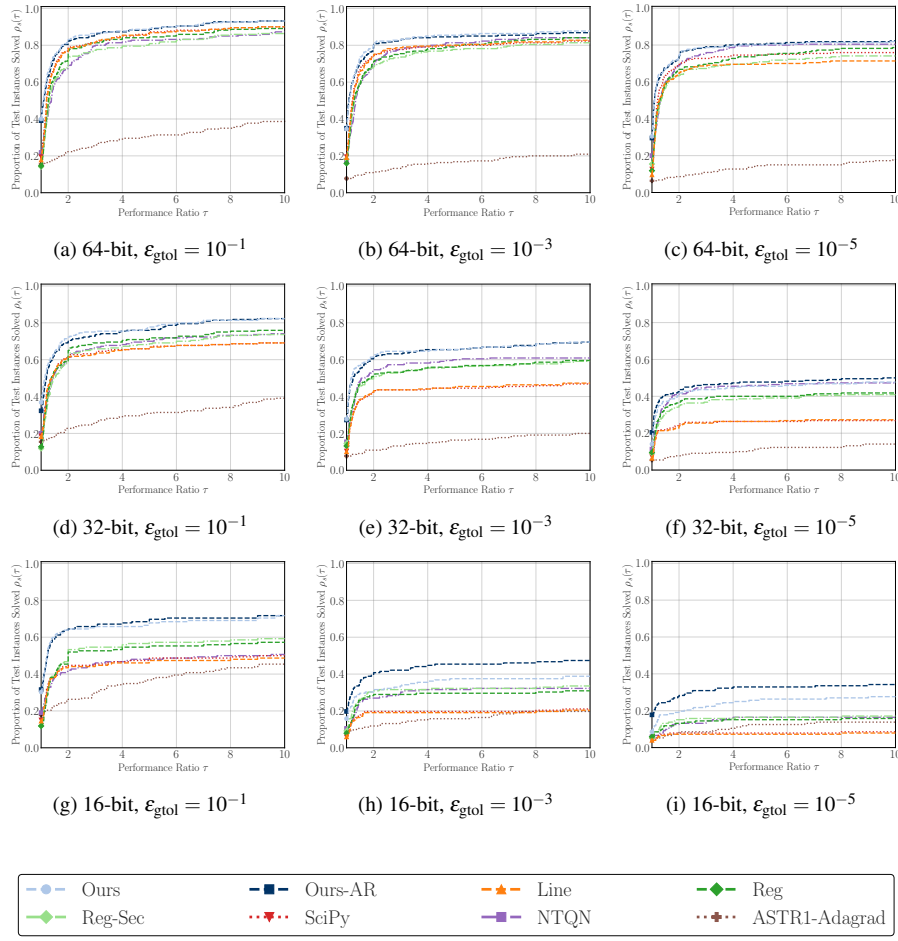


Fig. 4: Performance profiles across floating-point precisions (64-, 32-, and 16-bit) and tolerance levels.

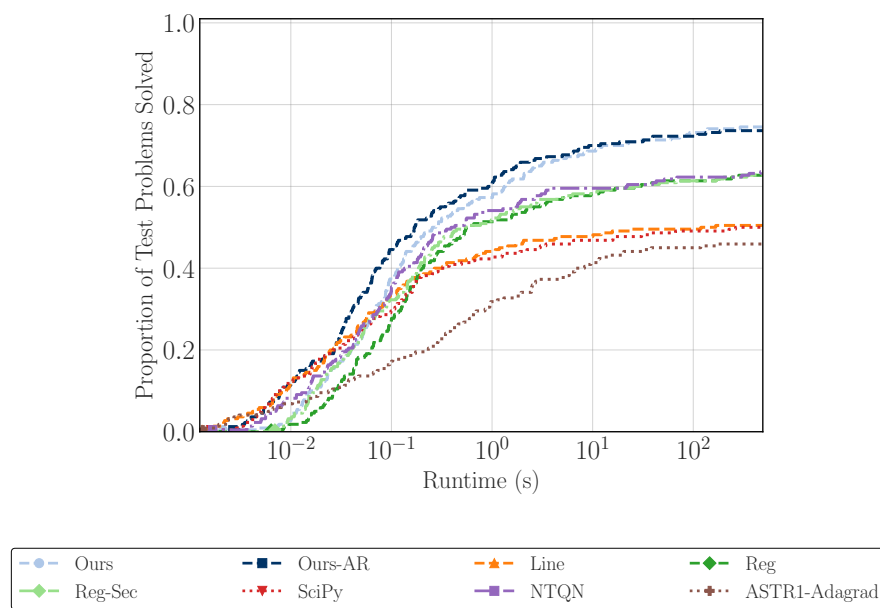


Fig. 5: Proportion of 32-bit test problems attaining $\varepsilon_{\text{gtol}} = 10^{-3}$ as a function of runtime.

7 Conclusion

We proposed a hybrid regularized quasi-Newton method for smooth nonconvex optimization with bounded errors in objective-function values. The method retains the order-optimal $\mathcal{O}(1/\varepsilon^2)$ first-order iteration complexity of the L -smooth nonconvex setting while selectively using noisy function values. At 16- and 32-bit precision, both proposed variants reached the strictest tested gradient tolerance on more CUTEst problems than the existing quasi-Newton baselines.

Future work includes the investigation of local convergence properties, the extension to constrained optimization, and the applications to practical problems in machine learning and scientific computing. Additionally, exploring adaptive strategies for selecting algorithmic parameters based on problem characteristics could further enhance the method's performance. An explicit analysis of the effect of inexact gradient evaluations would also provide valuable insight into the method's robustness and guide further enhancements.

A Proof of Lemma 6

The proof of Lemma 6 is similar to previous arguments [43, Lemma 3.2] and [17, Lemma 3.1].

Proof Recall that eq. (7) defines μ_k as $\theta_k \sqrt{\varsigma + \sum_{j \in K^+, j \leq k} \|g_j\|^2}$, and line 8 of Algorithm 1 asserts $\theta_k \in [\theta_{\min}, \theta_{\max}]$. Let $k_i \in K^+$ be the i -th element of K^+ . Define $S_{-1} := \varsigma$ and $S_i := \varsigma + \sum_{\ell=0}^i \|g_{k_\ell}\|^2$ for $i \geq 0$. For any $0 < x < y$, $x/y \leq -\ln(1 - x/y) = \ln y - \ln(y - x)$ holds. Thus, we have

$$\begin{aligned} \frac{\|g_{k_i}\|^2}{\mu_{k_i}^2} &\leq \frac{1}{\theta_{\min}^2} \frac{\|g_{k_i}\|^2}{S_i} && \text{(definition)} \\ &\leq \frac{1}{\theta_{\min}^2} (\ln S_i - \ln S_{i-1}). && (x/y \leq \ln y - \ln(y - x)) \end{aligned}$$

Similarly, $x/\sqrt{y} \geq x/(\sqrt{y} + \sqrt{y-x}) = \sqrt{y} - \sqrt{y-x}$ holds. Thus, we have

$$\begin{aligned} \frac{\|g_{k_i}\|^2}{\mu_{k_i}} &\geq \frac{1}{\theta_{\max}} \frac{\|g_{k_i}\|^2}{\sqrt{S_i}} && \text{(definition)} \\ &\geq \frac{1}{\theta_{\max}} (\sqrt{S_i} - \sqrt{S_{i-1}}). && (x/\sqrt{y} \geq \sqrt{y} - \sqrt{y-x}) \end{aligned}$$

Summing these inequalities over $0 \leq i < |K^+|$ yields the claim. $\square$

B Proof of Proposition 2

The proof of Proposition 2 uses the standard BFGS update.

Proof From eq. (31), the initial scalar satisfies $0 < \gamma = \left\| y_{k_{p-1}} \right\|^2 / (y_{k_{p-1}}^\top s_{k_{p-1}}) \leq \Lambda$. Let B and B_+ denote the positive definite Hessian approximations before and after one update. The BFGS update rule is

$$B_+ = B - \frac{B s s^\top B}{s^\top B s} + \frac{y y^\top}{y^\top s}.$$

Positive definiteness follows from $B \succ 0$, $s \neq 0$, and $y^\top s > 0$. Moreover, eq. (31) and the fact that $B s s^\top B / (s^\top B s)$ is positive semidefinite give

$$B_+ \preceq B + \frac{y y^\top}{y^\top s} \preceq B + \Lambda I.$$

Applying this inequality recursively over p updates from $B^{(0)} = \gamma I \preceq \Lambda I$ proves eq. (32). $\square$

The fixed-initial-scale upper-curvature inequality is enforced directly by eq. (33), while the fixed shift in eq. (34) supplies the lower spectral bound. No convexity or cocoercivity assumption is used. This point is important because the convergence theorem permits nonconvex objectives.

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Statements and Declarations

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Competing Interests

The authors have no relevant financial or non-financial interests to disclose.

Author Contributions

All authors contributed to the study conception and design. The first draft of the manuscript was written by Hiroki Hamaguchi, and all authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

Data Availability

All data analyzed during this study are publicly available. URLs are included in this article.

Code Availability

The code used in this study is publicly available at the following GitHub repository: <https://github.com/HirokiHamaguchi/qnlab>.